



Understanding Hallucinations

When AI confidently says things that are completely, entirely, dangerously wrong.



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What is **Hallucination**?



When a model generates output that is **factually incorrect**, **fabricated**, or **nonsensical** — yet delivered with full confidence, as if it were true.



The Core Problem

LLMs don't "know" facts. They predict statistically likely text — sometimes that output looks factual but isn't grounded in reality.



Why It's Dangerous

The model rarely signals uncertainty. Confident tone makes it hard for users to distinguish truth from fabrication.





In Plain English

Three everyday ways to picture what's happening.



The Overconfident Student

01

A student who skipped the chapter but answers every question with total confidence — sounding right, but making it up.



The Parrot That Sounds Smart

02

A parrot can recite Shakespeare flawlessly without understanding a word. Patterns aren't comprehension.



The GPS That Invents Roads

03

Imagine a GPS that, when unsure, confidently routes you down a road that simply doesn't exist.





Real-World Examples

These aren't theoretical — they've caused real damage.



The Lawyer Who Got Fined

In 2023, a NY lawyer cited 6 court cases ChatGPT invented. The judge fined him \$5,000 — none existed.



Books That Don't Exist

Ask for the 'best books on X' and you'll get plausible titles, authors, even publishers — none real.



Code That Looks Perfect

Models suggest API methods that look syntactically valid — but were never part of any real library.



Fake Quotes from Real People

LLMs attribute invented quotes to famous figures — phrased in their style, with a fabricated source.



Why do LLMs Hallucinate?



Training Data Noise

Web-scraped data contains misinformation, contradictions, and outdated content — the model learns these patterns too.



Probabilistic Generation

LLMs sample from probability distributions over tokens. High-temperature sampling increases creative and incorrect outputs.



Knowledge Gaps

When asked about topics underrepresented in training, models fill the void with plausible-sounding fabrications.



Sycophancy & RLHF

Models trained to please humans may generate confident, agreeable responses rather than uncertain accurate ones.





When LLMs Hallucinate Most

Stay extra skeptical when the topic falls in any of these:



Specific Numbers & Stats



Citations & Sources



Events Past Cutoff



Niche Expert Domains



Long-Form Generation



Multi-Step Math





4 Types of Hallucination



01

Factual Hallucination

Model asserts false real-world facts — fabricated citations, wrong dates, non-existent people.

"Einstein won Nobel in 1925" × (it was 1921)



02

Faithfulness Hallucination

Output contradicts or adds to the provided context — claims something not in the source document.

Summary with facts not in the original



03

Logical Hallucination

Reasoning steps look plausible but arrive at wrong conclusions through flawed inference.

Correct premises — wrong deduction



04

Temporal Hallucination

Model applies outdated information as if current — ignoring knowledge cutoff limitations.

Treating a 2022 CEO as still in role





How to Detect Hallucinations



1 Cross-Reference Verification

Verify specific facts, citations, names, and statistics against authoritative primary sources.



2 Consistency Sampling

Ask the same question multiple times with varied phrasing. Hallucinations often vary wildly.



3 Ask for Sources & Reasoning

Prompt the model to cite sources and explain step-by-step. Fabrications break down under scrutiny.



4 Automated LLM-as-Judge

Use a separate model to score outputs for factual consistency using NLI or SelfCheckGPT.



5 Calibrated Uncertainty Prompts

Ask the model to rate its own confidence. Well-calibrated models express genuine uncertainty.





How to Prevent Hallucinations



Use RAG (Retrieval-Augmented Generation)

Give the model your own documents to read from. Answers grounded in your data are harder to fabricate.



Lower the Temperature

Set temperature to 0–0.2 for factual tasks. Higher values mean more creativity and more invention.



Provide Clear Context

State what you know, what you want, and what's off-limits. Ambiguity invites fabrication.



Use Few-Shot Examples

Show the model 2–3 correctly answered questions before asking yours. Patterns guide accuracy.



Make 'I Don't Know' a Valid Answer

Explicitly tell the model it's okay to say it doesn't know. Most won't admit it unless invited.





Key Takeaways

Five things to keep in mind every time you use an LLM.



LLMs predict text — they don't verify facts.

01



Confident tone ≠ accurate answer.

02



Always verify critical facts against primary sources.

03



Use RAG and low temperature for serious work.

04



Treat AI as a junior collaborator, not an oracle.

05



What Is Caching?

Store the result of an expensive operation once,
then reuse it every time after.

THE SIMPLE IDEA

Do it once → reuse many times

First Time

User asks "What is machine learning?" → LLM generates
answer → **stored**

Next Time

Same question asked → **return stored response instantly** —
zero LLM call



ENJOYED THIS BREAKDOWN?

Share it with your team

One repost could save a colleague hours
spent re-learning caching from scratch.

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